战队名称

哈哈哈

比赛排名

总分数：1173

学生组排名：34

总排名：46

战队成员

哈哈哈、李方朔

哈哈大王、李易澄

马桶骑士、韩健坤

签到

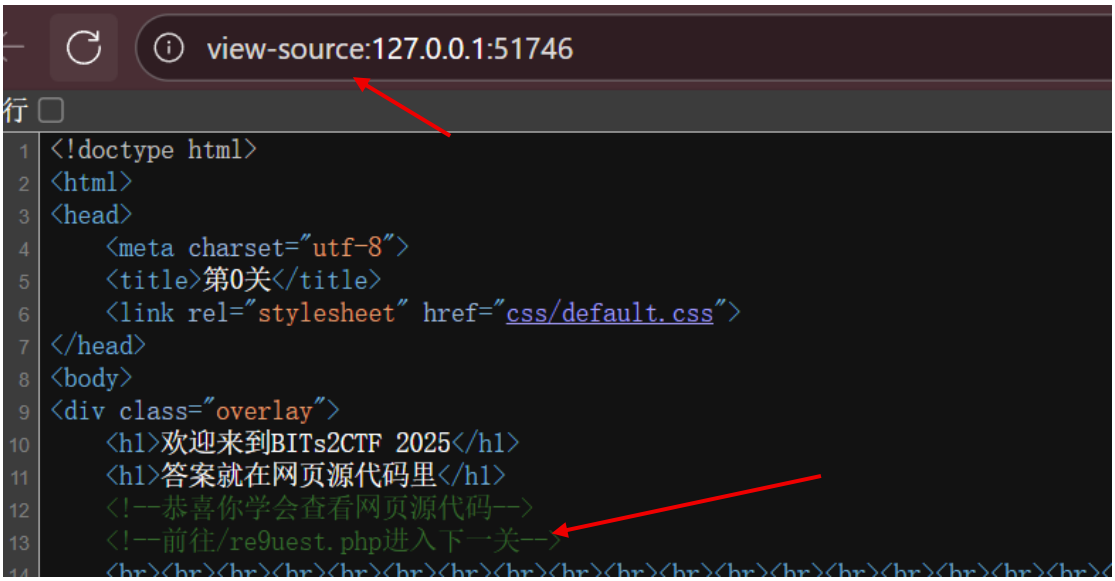
1、跳转网页

2、上传图片

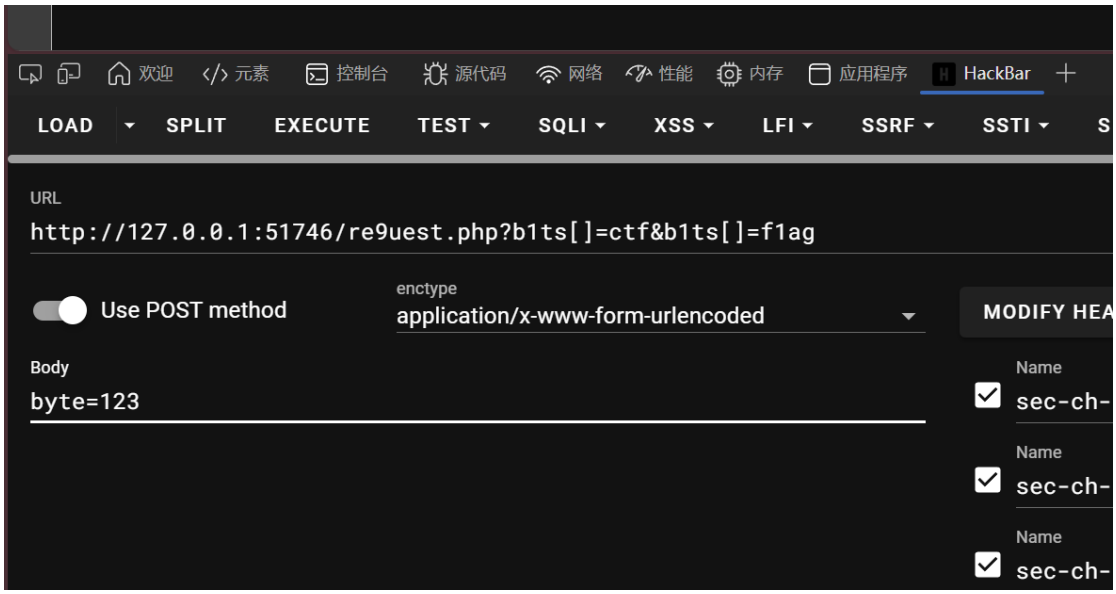
3、拿到 flag

Http

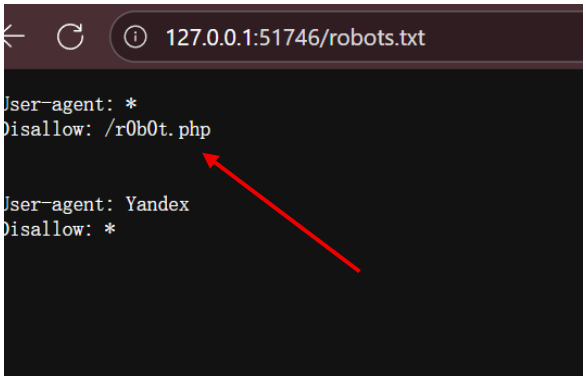
1、跳转网站，提示要看源代码



2、进入第一关是传参数，可以用 hackbar

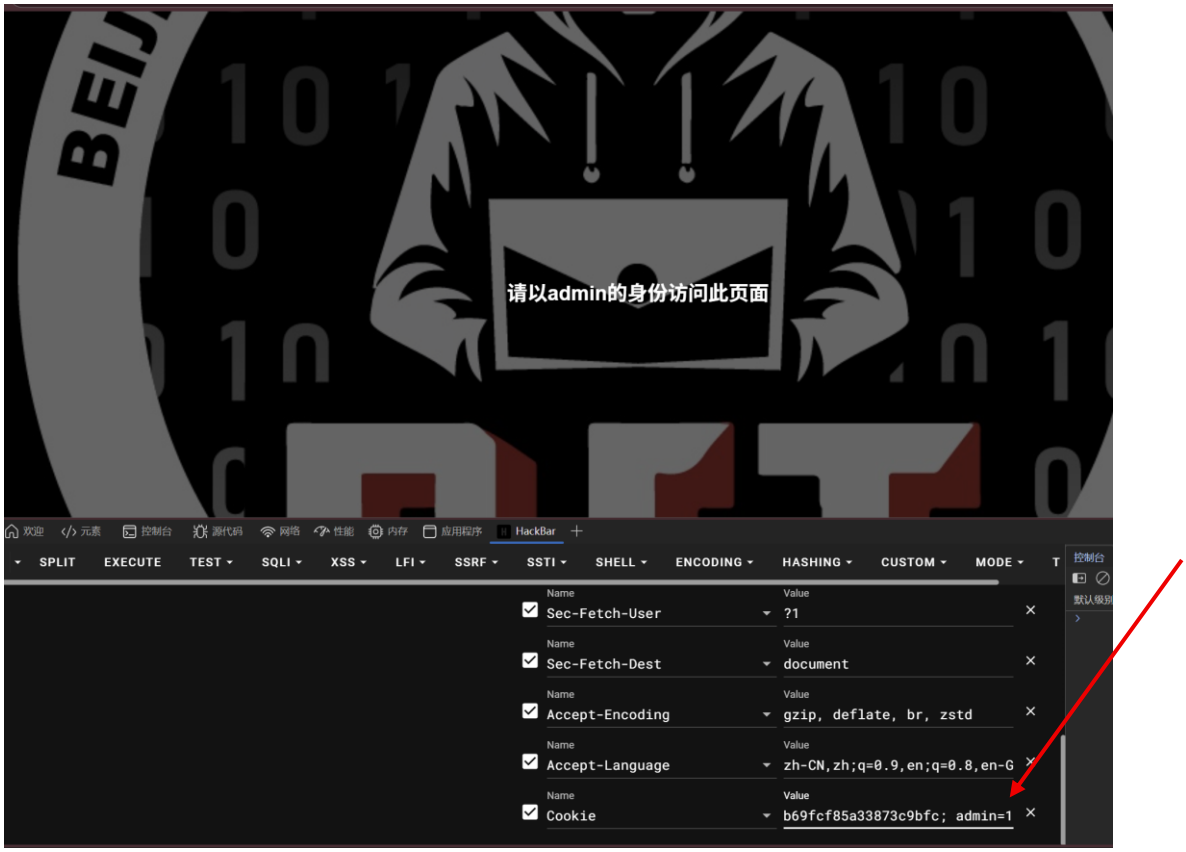


3、进入第二关，提示限制搜索引擎的爬取范围，大多数网站都会定义 robots.txt 文件来限制爬虫，打开之后发现这个文件



4、打开以后发现限制本地访问，改一下 X-Forwarded-For 为 127.0.0.1，提示进入下一关

5、要求身份，把 cookie 改一下



6、这样就拿到了 PD9waHAgJ0JJVHMyQ1RGezA2NjgxMWIxLWQxNWMtNGE4NS04NzUwLThjNmQxYjcxODVhYX0nID8+，base64 编码解码即可。

ChaCha20

1.首先用一下简单的线上反编码工具得到解码之后的原加密代码

python 反编译工具

请选择pyc文件进行解密，支持所有Python版本

选择文件 未选择文件

```
#!/usr/bin/env python
# visit https://tool.lu/pyc/ for more information
# Version: Python 3.10

from cryptography.hazmat.primitives.ciphers import Cipher, algorithms
from cryptography.hazmat.backends import default_backend
from Crypto.Util.number import long_to_bytes, bytes_to_long
from secret import flag
import time
import random
import os

def chacha20_encrypt(plaintext = None, key = None, nonce = None):
    cipher = Cipher(algorithms.ChaCha20(key, nonce), None, default_backend(), **('mode', 'backend'))
    encryptor = cipher.encryptor()
    ciphertext = encryptor.update(plaintext)
    return ciphertext

if __name__ == '__main__':
    k1 = bytes_to_long(os.urandom(32))
    n1 = bytes_to_long(os.urandom(16))
    print(f'''k1={k1}''')
```

代码如下：#!/usr/bin/env python

visit https://tool.lu/pyc/ for more information

Version: Python 3.1

from cryptography.hazmat.primitives.ciphers import Cipher, algorithms

from cryptography.hazmat.backends import default_backend

from Crypto.Util.number import long_to_bytes, bytes_to_long

from secret import flag

import time

import random

import os

def chacha20_encrypt(plaintext = None, key = None, nonce = None):

    cipher = Cipher(algorithms.ChaCha20(key, nonce), None, default_backend(), **('mode', 'backend'))

    encryptor = cipher.encryptor()

    ciphertext = encryptor.update(plaintext)

    return ciphertext

if __name__ == '__main__':

    k1 = bytes_to_long(os.urandom(32))

    n1 = bytes_to_long(os.urandom(16))

    print(f'''k1={k1}''')

    print(f'''n1={n1}''')

    random.seed(int(time.time()) % 100)

    k2 = random.getrandbits(128)

```
    n2 = random.getrandbits(64)
    key = long_to_bytes(k1 ^ k2)
    nonce = long_to_bytes(n1 ^ n2)
    encrypted_data = chacha20_encrypt(flag, key, nonce)
    print(f"加密后: {encrypted_data.hex()}")
    return None
```

2.接着使用所给的源代码还有 key1,nonce1,以及明文，使用 ai 工具

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得到能够输出所有种子的原始解密数据，最终在第 37 个种子处找到了 flagBITs2CTF{pyc_reverse_is_important}。